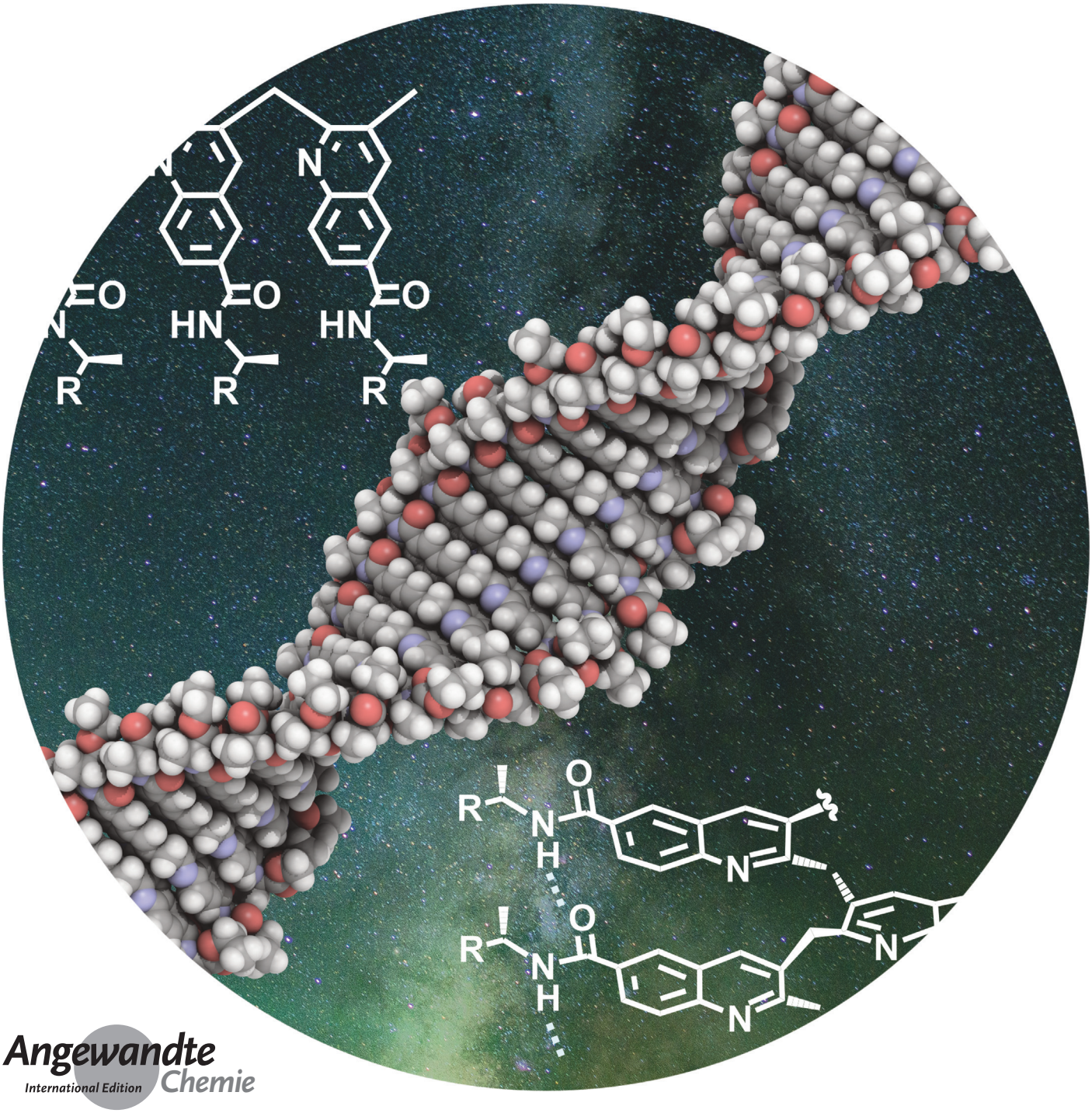
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Helical Polymers | Hot Paper

Construction of Helically Stacked π-Electron Systems in Poly(quinolylene-2,3-methylene) Stabilized by Intramolecular Hydrogen Bonds

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Abstract: *π-Stacked polymers, which consist of layered π-electron systems in a polymer, can be expected to be used in molecular electronic devices. However, the construction of a stable π-stacked structure in a polymer is considerably challenging because it requires sophisticated designs and precise synthetic methods. Herein, we present a novel π-stacked architecture based on poly(quinolylene-2,3-methylene) bearing alanine derivatives as the side chain, obtained through the living cyclo-copolymerization of an o-allenylaryl isocyanide. In the resulting polymer, the neighboring quinoline rings of the main chain form a layered structure with π–π interactions, which is stabilized by intramolecular hydrogen bonds. The vicinal quinoline units form two independent helices and the whole molecule is a twisted-tape structure. This structure is established on the basis of UV/CD spectra, theoretical calculations, and atomic-force microscopy.*

A DNA duplex is formed from a multilayered structure of an aromatic nucleobase pair via hydrogen bonds and π–π interactions. The layered π-electron systems exhibit unique opto-electric properties, including electron transport,[1] and DNA is regarded as an attractive model for molecular functional materials. To construct artificial layered π-electron systems, the construction of DNA-like polymers has advantages in that the π-conjugated molecules form a stable ordered structure along the polymer chain due to the covalent bonds, leading to very tight or close packing.[2] Therefore, π-stacked polymers have received considerable attention, and several types of polymers[3] and oligomers[4] have been synthesized. These studies are relevant to not only practical applications but also the investigation of the relationship between the physical properties and the polymer structures, including the molecular length[3b] or sequence.[4a,k,l] The general strategy for synthesizing π-stacked polymers is shown in Figure 1a: the planar π-electron systems are connected at regular intervals with appropriate linkers. However, it is not easy to synthesize a sophisticated structure. Most π-stacked polymers are based on condensation or coupling reactions of π-electron systems with linkers, and it is difficult to control the molar mass, molar-mass dispersity, and sequence, all of which are important factors that determine the properties of polymer materials. Therefore, novel concepts to construct π-stacked polymers are indispensable.

Recently, we achieved living cyclo-copolymerization of *o*-allenylphenyl isocyanide to afford a completely novel polymer architecture, poly(quinolylene-2,3-methylene) (poly-**1**),

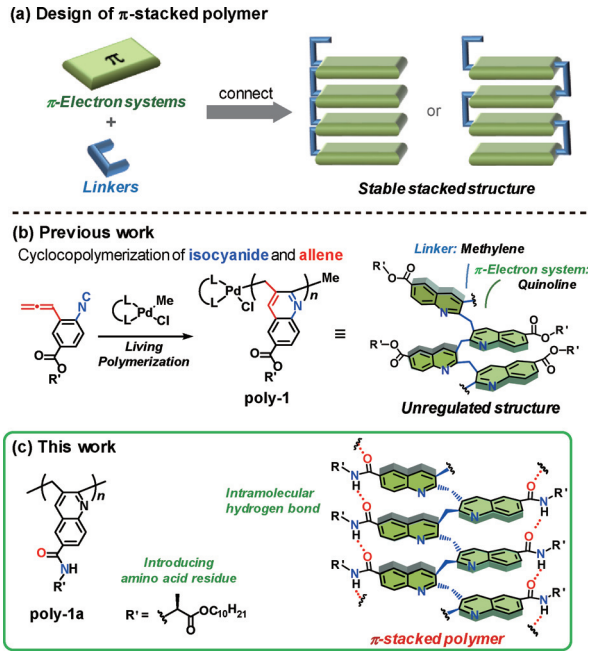


Figure 1. The concept of this study: poly(quinolylene-2,3-methylene) bearing an amide group to form the stable π-stacked structure.

whose molar mass and molar-mass dispersity can be controlled (Figure 1b).[5] The obtained poly-**1** consists of π-electron-system quinoline rings connected by methylene linkers, and a quinoline ring in poly-**1** can be in the vicinity of another quinoline ring located two residues apart within a distance that allows π–π stacking. This unique structure motivated us to explore the new type of π-stacked polymers. However, a stable stacked structure has not been observed for poly-**1** in solution, because the π–π interactions between neighboring quinoline rings are not strong enough to stabilize the multilayered structure. Herein, we describe the construction of a novel π-stacked polymer using poly(quinolylene-2,3-methylene) as a scaffold by introducing an alanine residue into the side chain (Figure 1c). In the polymer, the quinoline rings form helically multilayered structures with π–π interactions stabilized by hydrogen bonds between the neighboring units.

To realize the stable π-stacked structure of the quinoline rings in poly-**1**, monomer **1a** bearing an amide group derived from L-alanine was designed and synthesized. The hydrogen-bond chain between the neighboring amide residues allows the formation of the quinoline-layered structure in poly-**1a**, and the helicity is determined by the chirality of alanine.[6] According to the reported procedure,[5a] the polymerization of **1a** initiated by [1,3-bis(diphenylphosphino)propane]Pd-MeCl (**2**) in dichloromethane at different initial feed ratios ([**1a**]/[**2**] = 10, 20, 30, and 50) was conducted at 25 °C (Figure 2). In all cases, the polymerization proceeded with quantitative monomer conversion. After polymerization, the terminal palladium was removed via treatment with acetic acid to give quantitative yields of poly-$\mathbf{1a}_n$. The molar mass (M_n) and molar-mass dispersity were determined by size-exclusion chromatography (SEC) in THF using a polystyrene standard calibration. Since M_n exhibited a linear relationship

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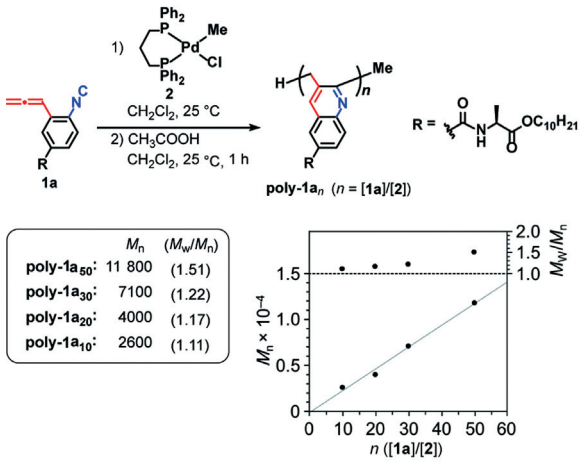


Figure 2. Cyclo-copolymerization of monomer **1a** with palladium initiator **2**, and plots of M_n and M_w/M_n of poly-**1a**$_n$ against the initial feed ratio of **1a** and **2**.

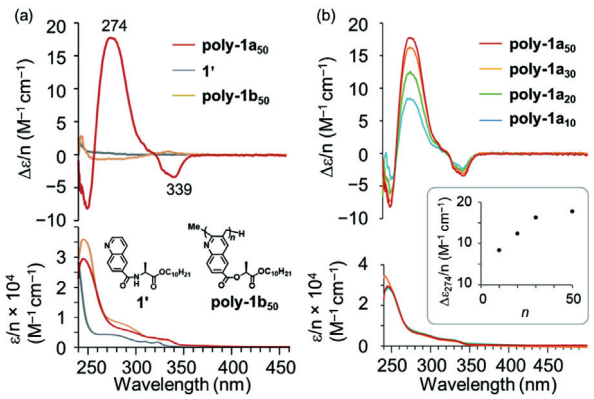


Figure 3. CD and UV spectra of a) poly-**1a**$_{50}$, poly-**1b**$_{50}$, and **1′**, and b) poly-**1a**$_n$ in $CHCl_3$ at 25 °C. Concentrations: [poly-**1a**$_{50}$] = 4.4×10^{-2} mM, [poly-**1b**$_{50}$] = 4.3×10^{-2} mM, [**1′**] = 4.0×10^{-2} mM, [poly-**1a**$_{30}$] = 4.3×10^{-2} mM, [poly-**1a**$_{20}$] = 4.3×10^{-2} mM, and [poly-**1a**$_{10}$] = 4.6×10^{-2} mM.

with the [**1a**]/[**2**] feed ratio, the polymerization of **1a** also has a living nature (Figure 2). However, the molar-mass dispersity became broader as M_n increased, probably due to the formation of a stable secondary structure, as will be described later. ^{1}H NMR spectroscopic measurements of poly-**1a**$_{50}$ were conducted in a mixture of $CDCl_3$ and [D_6]DMSO (1:1 v/v), where DMSO was added to inhibit the formation of hydrogen bonds between the amide side chains; the spectrum exhibited signals associated with the desired quinolylene-2,3-methylene backbone (Figure S5, Supporting Information).[7] All polymers showed a good solubility in organic solvents, such as chloroform, tetrahydrofuran, dimethylformamide, toluene, and ethyl acetate.

To investigate the polymer conformation of poly-**1a**, the dimer (**3**: $n=2$), trimer (**4**: $n=3$), and hexamer (**5**: $n=6$) were synthesized and analyzed by ^{1}H NMR spectroscopic measurements in $CDCl_3$ (Figure S7).[8] The ^{1}H NMR spectra of **5** showed downfield shifts of hydrogen-bonded amide protons. Additionally, anisochronous signals of methylene linkers which show AB patterns and a spreading of signals over a wide range of chemical shifts in the aromatic regions were observed, which were not seen in **3**. These results suggest that **5** forms a folded structure with π–π interactions.[4i]

The conformation of poly-**1a** was investigated by UV/Vis-absorption and circular-dichroism (CD) spectroscopy (Figure 3a). The CD spectra of poly-**1a**$_{50}$ in chloroform showed characteristic Cotton effects at 339 nm and 274 nm; these features were quite different from those of the corresponding monomeric unit **1′** and poly-**1b**$_{50}$ containing a chiral ester group as the side chain instead of the amide group of poly-**1a**$_{50}$. The intensity of the Cotton effect depends on the molar mass of poly-**1a**$_n$ (Figure 3b), and not on the concentration, which was tested in the range from 3.8×10^{-1} to 3.8×10^{-3} mM (Figure S10a).[9] These results suggest that poly-**1a** forms an optically active secondary structure without intermolecular aggregation at the concentration used in the CD measurements. The CD signals changed little when the initial feed ratio (n) exceeded 30, indicating that the secondary structure would be stable at $n>30$. The IR spectra of poly-**1a**$_{50}$ in chloroform (5 mM) indicated intramolecular hydrogen bonds between the amide side chains, since the absorption of the amide N–H and C=O stretching bands was observed at 3258 cm^{-1} and 1643 cm^{-1} respectively, which is lower than the corresponding bands of **1′** at 3433 cm^{-1} and 1661 cm^{-1} (Figure S9b). When DMSO was added to a solution of poly-**1a**$_{50}$ in chloroform, a significant decrease in the Cotton effect was observed in the CD spectrum (Figure 4a), and the intensity of the signals at 274 nm exhibited a non-linear relationship with the concentration of DMSO (Figure 4c). These results indicate that the intramolecular hydrogen bonds in poly-**1a**$_{50}$ cooperatively stabilize the secondary structure, which is supported by the fact that poly-**1b**$_{50}$ is CD-silent (Figure 3a). A solution of poly-**1a**$_{50}$ in chloroform excited at 330 nm exhibited fluorescence emissions at 380 nm and 520 nm, which are probably caused by π–π interactions of the quinoline rings, since no emission was observed from the monomeric quinoline compound **1′** (Figure S11). The original

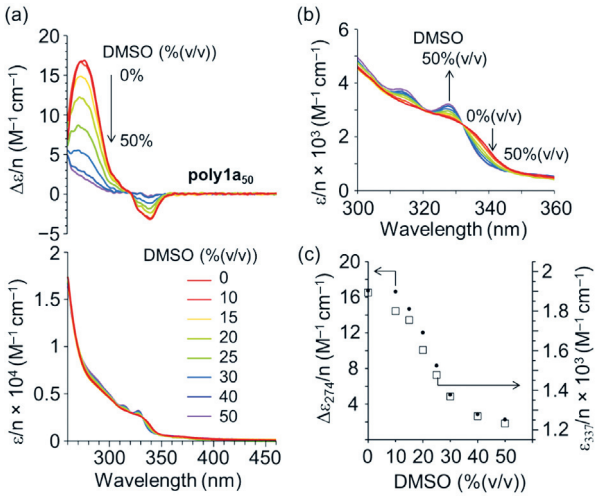


Figure 4. a) CD and UV spectra of poly-**1a**$_{50}$ in a mixture of $CHCl_3$ and DMSO at 25 °C, b) an enlarged view of the UV spectra in the range of 300–360 nm, and c) intensity of CD spectra at 274 nm (filled circles) and UV spectra at 337 nm (squares) against the ratio of DMSO in $CHCl_3$. Concentration: [poly-**1a**$_{50}$] = 4.7×10^{-2} mM.

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UV spectrum of poly-**1a**$_{50}$ in chloroform showed a hypochromic effect resulting in a broad signal, which disappeared upon the addition of DMSO as a competitive reagent for hydrogen bonds (Figure 4b). Such a hypochromic effect is observed when aromatic chromophores form a slip-stacked structure,[3b,10] which suggests that the quinoline units in poly-**1a** adopt a stable π-stacked structure in chloroform.

Based on these experimental results, we predicted the secondary structure of poly-**1a** in chloroform. An ideal poly(quinolylene-2,3-methylene) bearing an amide side chain derived from L-alanine methyl ester (poly-**1a**$_6$′) was constructed to form hydrogen bonds with the neighboring units (i and $i+2$), and a conformation search with energy minimization using MacroModel 11.9 with the OPLS3e force field[11] was conducted (Figure S12; for a detailed description of the calculations, see the Supporting Information).[12,13] To simulate the CD spectra, time-dependent density-functional-theory (TD-DFT) calculations were performed. For a minimum unit forming a π-stacked structure, a tetramer (poly-**1a**$_4$′) was used. The initial structure of poly-**1a**$_4$′ was created from the optimized structure of poly-**1a**$_6$′, as mentioned above, and re-optimized by DFT calculations using the B3LYP/6–31G** method. Subsequent TD-DFT calculations using the same method afforded reasonable UV and CD spectra. Although the calculated CD spectrum was not very accurate in the shorter-wavelength region, the calculated UV and CD spectra closely agreed with the observed spectra of poly-**1a** (Figure 5a). This result indicates that the structure of the calculated poly-**1a**$_4$′ is a reasonable local conformation of poly-**1a**. Based on the optimized structure of poly-**1a**$_4$′, the molecular structures of poly-**1a**$_{10}$′ and poly-**1a**$_{40}$′ were constructed and optimized by molecular-mechanics minimizations using MacroModel (Figure 5b,c). These optimized structures suggest that the amide side chains form hydrogen bonds between neighboring units, and that the polymer forms a right-handed (P) 2_1-like helix with a multilayered structure. During molecular-dynamics simulations at 300 K for 1 ns in chloroform, the layered structure and hydrogen bonds were maintained (Figure S15).[14] The calculated distance between the centers of the neighboring quinoline rings of poly-**1a**$_{10}$′ is 4.2 Å, and the distance between the faces of the quinoline rings is 3.2 Å, which falls within the range of π–π interactions (Figure 5b).[15] The vicinal quinoline units form two independent helices and the whole molecule forms a twisted-tape structure with a pitch of 40 Å (Figure 5c).

The helical structure of the poly(quinolylene-2,3-methylene)-bearing amino-acid residues was investigated by atomic-force microscopy (AFM). Figures 6 and S17 show the AFM images of poly-**1a**$(C_{18})_{50}$, which was prepared from a monomer containing an alanine residue with an octadecyl ester to promote crystallization through the long alkyl chain. A spin-cast film was made from a diluted chloroform solution (0.01 mg mL^{-1}) on highly oriented pyrolytic graphite (HOPG), which was annealed at 50 °C for 8 h. The individual polymer chains could be clearly resolved and a uniform helical morphology with a pitch of 3.6 ± 0.4 nm was found in each fiber. This result suggests that the proposed structure obtained by the above computational simulations is reasonable.

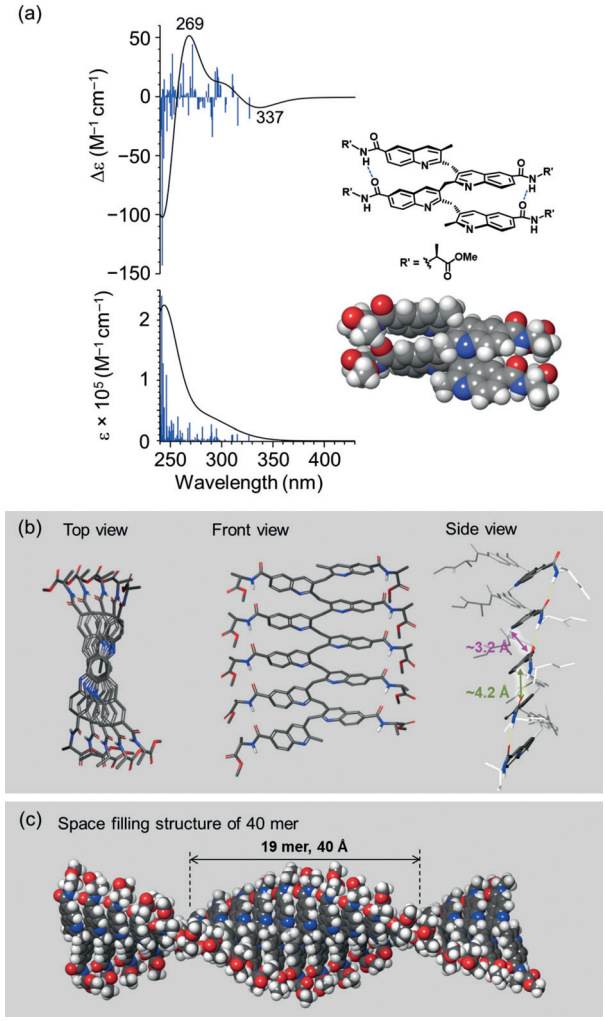


Figure 5. a) Simulated CD/UV spectra of a reasonable poly-**1a**$_4$′ structure using TD-DFT. Schematic and space-filling structures are also shown (right). b) Proposed structure of poly-**1a**$_{10}$′ obtained through molecular-mechanics optimization. In the side view, the layered quinoline units and amide side chain forming hydrogen bonds are emphasized. c) Space-filling structure of poly-**1a**$_{40}$′.

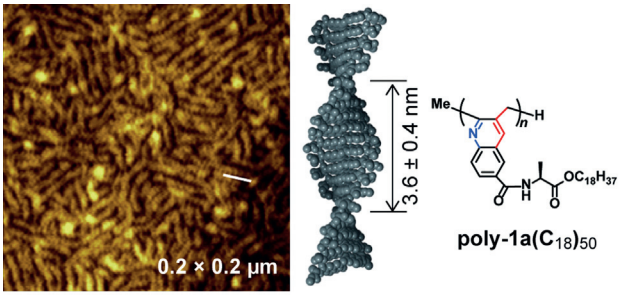


Figure 6. AFM image of poly-**1a**$(C_{18})_{50}$ on HOPG. A schematic representation of the right-handed helical polymer is also shown.

In conclusion, we have successfully synthesized a novel stacked π-electron system: poly(quinolylene-2,3-methylene). The quinoline backbones in the polymer forms a 2_1-like helically folding structure with π–π interactions stabilized by two outer N–H···O=C hydrogen-bond chains, and the whole

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molecule is a twisted-tape structure. The spectroscopic data, computational calculations, and AFM images strongly suggest the proposed structure. The polymer is well soluble in common organic solvents, and the stacked structure can be made disordered by adding a hydrogen-bond-competitive reagent, DMSO. The presented novel π-stacked architecture cannot be synthesized using conventional organic synthesis and polymerization, and this 2_1-helical conformation has only been known for limited types of polymers[3j,k] and some polydienes in the solid state.[16] Additionally, this polymer is prepared by living polymerization of **1**, and further modifications of the molecular design, such as the monomer sequence and end-structure, are possible. Also, the resulting polymer has an optically active helical structure, suggesting a potential use in chiral opto-electronic devices. Further research on the synthetic methods and the optical as well as electrochemical properties of the polymer is in progress.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenes · cyclo-copolymerization · helical polymers · isocyanide · π-stacked polymers

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[7] Unfortunately, the signal of the terminal methyl group was not clearly observed in the ^{1}H NMR measurements due to overlap with DMSO, and the absolute M_n could not be calculated from the spectrum.

[8] When a monomer containing an alanine residue with a methyl ester was used, partial association occurred. In this experiment, a monomer containing a leucine residue with a methyl ester was used (for details, see the Supporting Information).

[9] Other CD spectra of poly-**1a**$_{50}$ confirming the temperature and solvent influences are shown in Figure S10.

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[13] The dipole moments of poly-**1a**$_1$′ and poly-**1a**$_4$′ were calculated using DFT (see Figure S14). In the case of poly-**1a**$_4$′, the dipole

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moments of the local units cancelled with the vicinal units (i and $i\pm1$), and the dipole moment derived from translational axis was conserved.

[14] In the case of poly-$\mathbf{1b}_{10'}$, which was built by replacing the amide nitrogens in poly-$\mathbf{1a}_{10'}$ with oxygens, followed by a molecular-mechanics minimization, the layered structure became completely disordered in the molecular-dynamics simulation (Figure S16).

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